Alupana Balaji Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.99
A.P Model School & Jr College, Tamada	Class XII (Board of Intermediate Education Andhra Pradesh (BIEAP))	2021	94.70%
A.P Model School & Jr College, Tamada	Class X (Board of Secondary Education Andhra Pradesh (BSEAP))	2019	9.50%

SCHOLASTIC ACHIEVEMENTS

- **1st Prize - District-Level Competition** -- Rashtriya Madhyamik Shiksha Abhiyan, Literary & Cultural Competitions, **2018**
- **1st Rank - Srikakulam District & State-Level Participant** -- Vidyarthi Vigyan Manthan (VVM), **2018-19**

Competitions & Hackathons

- **Team Lead -- Discoverathon 2026, Novartis** | Led a **4-member team** to develop an ML-based solution for **predicting drug combination activity**.
- **College Representative -- Indian International Science Festival (IISF)**, Kolkata, **2019**.

PROJECTS

Patient Readmission Analysis | Python, SQL, MySQL, Power BI

- Analyzed **101K+ hospital encounters** to identify factors associated with 30-day readmission.
- Performed **data cleaning, feature engineering, and statistical analysis** across key readmission factors.
- Used **SQL/MySQL** to uncover patterns across diagnoses, medications, and healthcare utilization.
- Built an **interactive Power BI dashboard** highlighting readmission rates from **8.44% to 25.66%** with prior inpatient visits.

Production of Protease from Soil Actinomycetes under Submerged Fermentation using Mixed Substrate of Barley Powder and Corn Flour

Andhra University College of Engineering, Visakhapatnam | Nov 2024 - Apr 2025

- Isolated and screened **protease-producing Actinomycetes** from soil using selective culturing and screening assays.
- Optimized **submerged fermentation** using barley powder and corn flour as low-cost substrates.
- Quantified **protease activity** using biochemical assays and evaluated key fermentation parameters.

TECHNICAL SKILLS

Languages: Python | SQL

CS Fundamentals: Data Structures & Algorithms | Object-Oriented Programming | DBMS

Software Development: REST APIs | Debugging | Git | GitHub

Python & Data: NumPy | Pandas | Matplotlib

Tools: VS Code | Jupyter Notebook

Soft Skills: Communication | Teamwork & Collaboration | Leadership & Coordination | Problem Solving | Analytical Thinking | Time Management | Adaptability

POSITIONS OF RESPONSIBILITY

Placement Coordinator | IIT Hyderabad

M.Tech Medical Biotechnology | *May 2026 - Present*

- Coordinate placement-related activities, student communication, and engagement with the institute's placement process.

Teaching Assistant | IIT Hyderabad

M.Tech Medical Biotechnology | *July 2026 - Present*

- Support course delivery and academic activities, including student coordination and course-related assistance.

Class Representative | Andhra University College of Engineering, Visakhapatnam

B.Tech Biotechnology | *June 2022 - May 2025*

- Represented the class and facilitated communication and coordination between students and faculty.

WORKSHOPS, CONFERENCES & PRESENTATIONS

WORKSHOPS & CONFERENCES

- **Participant -- 7th Bangalore Cognition Workshop (BCW 2026)**, Centre for Neuroscience, IISc Bengaluru | June-July 2026.
- **Participant -- International Workshop on Sustainable Practices for Industrial Pollution Control**, Andhra University College of Engineering, Visakhapatnam.

POSTER PRESENTATIONS

- **"Unveiling the Molecular Ballet: Computational Exploration of MCL1 Protein Interactions for Leukaemia Therapeutics"** -- Poster Presenter, **Andhra Bio Summit 2024**.

EXTRACURRICULAR

- **Gold Medal -- Relay**, Diesta, Inter-Departmental Sports & Cultural Fest, IIT Hyderabad
- **Silver Medal -- 100 m Athletics**, BioBlast, Annual Techno-Cultural & Biotechnology Event
- **Silver Medal -- Carroms**, BioBlast, Annual Techno-Cultural & Biotechnology Event
- **Hobbies & Interests:** Chess | Badminton | Running & Athletics | Exploring Music